

Formation of the Sedna Sphere in the Solar Natal Embedded Star Cluster (Brasser, Duncan, & Levison 2006)

The Sedna Sphere (Inner Oort Cloud)

$a \in [1000, 15000] \text{ AU}$, $M_{\text{IOC}} \approx 4 - 12 M_{\oplus}$

Dominant Reservoir ($M_{\text{IOC}}/M_{\text{OOC}} \approx 3 - 10$)

Perturbing Cluster Star
($M_* \approx 0.5 - 0.8 M_{\odot}$, $v_{\infty} \approx 1 \text{ km/s}$)

Impact Parameter
 $b \sim 500 - 1500 \text{ AU}$

Lifted Sedna Orbit (Trapped)

$q_{\text{new}} = 76.2 \text{ AU} > q_{\text{crit}}$

Decoupled from Neptune!

Transverse Tidal Kick

$\Delta v_{\perp} \sim 200 - 500 \text{ m/s}$

$\Delta J \approx r_a \Delta v_{\perp}$

Comet at Aphelion
($r_a \approx 2a \approx 1000 \text{ AU}$)

Initial Neptune-Scattered Orbit
($q_0 \approx 30 \text{ AU}$, $a \approx 500 \text{ AU}$)

Sun

Neptune
(30 AU)

Brasser et al. (2006) 3-Step Decoupling Mechanism:

1. Giant planets scatter comets outwards to $a \sim 10^2 - 10^4 \text{ AU}$.
2. Embedded cluster stellar flybys deliver tidal kicks at aphelion.
3. Perihelion lifted ($q > 36 \text{ AU}$), permanently insulating Sedna sphere.